

SHEHAN ROSHAN JAMAL

Software Engineer | Auckland, New Zealand

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Work Rights: Eligible for a New Zealand Post Study Work Visa (up to 3 years, open work rights) — application in progress. No employer sponsorship or accreditation required.

PROFESSIONAL SUMMARY

Software engineer with a Master of Software Engineering and close to four years of experience building and delivering full-stack systems for enterprise and research clients across Poland, the Netherlands, Australia, the US and New Zealand. Owned a geospatial research platform end to end — from requirements analysis and schema design through to production — and built an explainable-AI prediction system achieving 92% accuracy. Strong across Python, TypeScript, React/Next.js, PostgreSQL and AWS, with a track record of taking systems from design documentation through to deployment.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Go, SQL, PL/SQL

Frontend: React, Next.js, Leaflet, Electron, HTML/CSS

Backend & APIs: FastAPI, Node.js, REST, GraphQL, Apollo, OpenAPI 3.0, Auth0

Databases: PostgreSQL, PostGIS, MySQL, Supabase

Cloud & DevOps: AWS, Azure, Docker, GitHub Actions, CI/CD, Git

Machine Learning: CatBoost, scikit-learn, TreeSHAP, BERT, LSTM, OpenCV, Tesseract OCR

Tools & Practices: Jira, Agile delivery, technical specification writing, API design

PROFESSIONAL EXPERIENCE

Software Engineer (Internship) | Tiaki Taonga Trust, Auckland, New Zealand

April 2026 – July 2026 | Portfolio Owner, Geo-Spatial & Policy Research Platform

Tech: TypeScript, React, Next.js, PostgreSQL, PostGIS, AWS, Supabase, Leaflet, GitHub Actions, Jira

- Owned the Geo-Spatial & Policy Research Platform end to end as Portfolio Owner, taking it from requirements analysis through database schema design, API design and implementation to production deployment across a 15-week engagement.
- Designed a PostgreSQL/PostGIS schema on a hub-and-spoke model that made orphaned records structurally impossible, enforcing integrity through foreign-key and CHECK constraints and audit triggers that produced an immutable change log.
- Built 25+ REST endpoints in TypeScript with fully parameterised SQL, a standardised response envelope, custom domain error codes and pagination, documented in a published OpenAPI 3.0 specification.
- Built the React/Next.js frontend with Leaflet interactive mapping and a 7-role RBAC system, extended by a delegated per-project permission model enabling granular access control without code changes.
- Led cross-portfolio authentication integration into a shared monorepo, resolving an identity mismatch between two independently built systems while leaving all 25+ downstream endpoints unmodified.
- Migrated 19 API routes from Supabase PostgREST to raw parameterised SQL on AWS PostgreSQL, diagnosing spatial-function resolution, geometry serialisation and connection-pooling defects during the cutover.
- Presented live progress demonstrations throughout the engagement and a final end-to-end system demonstration to the Trust's technical and business stakeholders.
- Established CI/CD with GitHub Actions and versioned idempotent SQL migrations delivered via pull request, and produced full handover documentation for continued development.

Software Engineer | IFS, Sri Lanka

01/2022 – 01/2025 | 2 years 9 months

Tech: PL/SQL, SQL, XML, Azure, Jira

- Delivered customisations and new functionality for enterprise customers across Poland, the Netherlands, Australia and the US, working across IFS versions from App09 through Cloud 24R1.
- Advised Polish and Australian customers on PSO and related modules as functional advisor, recommending alternative implementation approaches that reduced the build effort of delivered customisations.

- Managed end-to-end version uplifts (App09 to App10, App10 to Cloud), covering the code, configuration and data changes each migration required.
- Developed across the application stack — PSO, dialogs, assistants, custom pages and bespoke functionality built to customer specification.
- Built customer-specific reports from scratch and modified existing XML report definitions to introduce new fields and behaviour for individual customer requirements.
- Delivered application localisation into Polish and French, and authored technical specifications and time-estimation sheets used by other developers for project planning.
- Built and remediated Tar scripts for environment data setup, adapting legacy scripts to newer application versions where required data structures had changed.
- Ran internal knowledge-sharing sessions onboarding new team members onto core application functionality.

Software Engineer (Internship) | Agile Financials, Sri Lanka

06/2020 – 06/2021 | 1 year

Tech: Go, React, GraphQL, PostgreSQL, Auth0

- Built the full sign-up and login flow end to end — React frontend, Go backend and PostgreSQL database — with Auth0 integration and email confirmation.
- Developed and tested the GraphQL API and Go backend, writing both integration and unit test coverage.
- Sourced historical market data from the Colombo Stock Exchange for the project's model and presented the finished product to National Trust Bank board members.

KEY PROJECTS

TFT-Predict v3 — Explainable AI Prediction System | Master's Capstone

2026 | Python, CatBoost, TreeSHAP, FastAPI, Electron, OpenCV

- Built a real-time prediction system achieving 92.08% Top-4 accuracy and 0.9798 AUC, using a screenshot-only pipeline that fed OCR-extracted game state through 23 engineered features into a CatBoost model.
- Raised trait recognition accuracy from 19.8% to 61.0% by diagnosing and fixing an identifier mismatch between display names and internal game data IDs.
- Delivered model transparency via TreeSHAP, generating human-readable explanations for every prediction and surfacing them through a FastAPI backend and Electron desktop overlay.

ReachOut — Volunteer Coordination Platform | Personal Project

2025 | Python, React, Node.js, Apollo, MySQL, Docker, OpenAI API, AssemblyAI

- Cut task creation time from 3 minutes to 30 seconds by integrating OpenAI and AssemblyAI to provide voice-to-text task entry.
- Architected and containerised the full stack — Python backend, React frontend and Apollo middleware — deployed with Docker.

Novel-to-Comic Generation Pipeline | Personal Project

2026 | FLUX.1, PuLID, Llama 3.1, ComfyUI, Vast.ai

- Built an end-to-end pipeline converting novel text into illustrated comic panels on GPU cloud infrastructure, combining FLUX.1 Dev for image generation, PuLID Flux II for character consistency and Llama 3.1 8B for scene parsing.

Emotion Analyzer for Articles | Research Project

2022 – 2023 | Python, BERT, LSTM

- Built a Chrome extension and Python API backed by a hybrid BERT and LSTM model that scored news articles on an emotional positivity scale.

EDUCATION

Master of Software Engineering | Yoobee College of Creative Innovation, New Zealand

2026

BEng (Hons) Software Engineering | Informatics Institute of Technology, Sri Lanka

Affiliated with the University of Westminster, UK

CERTIFICATIONS, AWARDS & PUBLICATIONS

AWS Cloud Foundation and AWS Cloud Security Fundamentals — AWS Academy | Cutting Edge 2018, Informatics Institute of Technology — 1st Place (Project) and 1st Place (Creativity)

